

Soohwan Lee

E-mail : soohwanlee@unist.ac.kr

 Personal Website |  Google Scholar |  LinkedIn |  GitHub |  Medium

Human-centered AI | Human-AI Collaboration | CSCW | Conversational AI | Social Psychology | Group Dynamics

RESEARCH STATEMENT

My research investigates how AI-mediated systems shape group dynamics and social influence beyond the individual level. Drawing on social psychological theories, I understand, design, and empirically study interactive AI systems that intervene in group processes to enhance collective intelligence, improve decision-making quality, and foster creative collaboration.

I advance the concept of Group-Centered AI, reframing user experience as collective and emergent rather than an aggregation of individual interactions. AI has the potential to fundamentally reshape social dynamics, and understanding this influence can enable the design of group processes that foster greater inclusivity, diversity, and systems-level thinking. My work aims to inform responsible AI systems that influence groups transparently and ethically while preserving critical thinking and agency.

EDUCATION

- **Ulsan National Institute of Science and Technology (UNIST)** Ulsan, South Korea
Ph.D in Design with Concentration in HCI (Expected) 2027
 - Dissertation(Tentative Title) - Understanding and Designing AI Systems that Reshape Social Dynamics in Group Decision-Making
 - Committee: Kyungho Lee (Chair), Dajung Kim, Seongkook Heo, Chi-Lan Yang, Hajin Lim
- **M.S in Creative Design Engineering with Concentration in HCI** 2022
 - Thesis - NEAS:AI-based Interactive System to Support Non-verbal Expressions in Online Communication
 - Committee: Kyungho Lee (Chair), Ian Oakley, Young-woo Park
- **B.S in Human Factors Engineering & Computer Science and Engineering** 2020

EXPERIENCE

- **Ulsan National Institute of Science and Technology (UNIST)** Ulsan, South Korea
Research Assistant 03/2020 – Present
 - Expressive Computing Lab (PI: Kyungho Lee)
Research Intern 12/2018 – 12/2019
 - Interactions Lab (PI: Ian Oakley)
- **University of Illinois Urbana-Champaign (UIUC)** Champaign, IL, United States
Entrepreneurship & Data Science Summer Camp 07/2019
 - Data Analysis and Consulting for Start-ups at UIUC

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=SHORT PAPER AND POSTER, W=WORKSHOP, T=THESIS, D=DOMESTIC, U=UNDER REVIEW

[Peer-reviewed Conference Proceedings]

- [C.1] **Soohwan Lee** and Kyungho Lee. (2026). **Understanding Compliance and Conversion Dynamics in Multi-Agent Collectives**. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI'26)
- [C.2] Seoyeong Hwang, Soohyun Hwang, **Soohwan Lee**, Dajung Kim and Kyungho Lee. (2026). **Augmentary: Exploring How LLM-Generated Interpretive Feedback Supports Meaning-Making in Reflective Journaling**. In Proceedings of the 2026 ACM Designing Interactive Systems Conference (DIS'26)
- [C.3] **Soohwan Lee**, Seoyeong Hwang, Ian Oakley, and Kyungho Lee. (2024). **Expanding the Design Space of Vision-based Interactive Systems for Group Dance Practice**. In Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS'24)
- [C.4] DoYoung Lee, **Soohwan Lee**, and Ian Oakley. (2020). **Nailz: Sensing Hand Input with Touch Sensitive Nails**. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)

[Journal Publications]

- [J.1] **Soohwan Lee**, Seoyeong Hwang, Mingyu Kim, Dajung Kim, and Kyungho Lee. (2026). **Investigating LLM-Powered Minority Support in Power-Imbalanced Group Decision-Making: Counterargument and Mediation as Intervention Strategies**. In Proceedings of the ACM on Human-Computer Interaction (CSCW'26)
- [J.2] Sangsu Jang, **Soohwan Lee**, Temirlan Dzhoroev, Taeyoon Kim, Hyejin Oh, Nari Kim, and Young-Woo Park (2021). **Design Guidelines for Contextual Awareness and Management of Hygiene in Daily Life with Infectious Viruses**. Archives of Design Research (Volume: 34 Issue: 3)

[Short Papers and Posters]

- [S.1] **Soohwan Lee** (2026). **Understanding and Designing AI Systems that Reshape Social Dynamics in Group Decision-Making**. In Companion Proceedings of the 31th International Conference on Intelligent User Interfaces (IUI '26 Companion)
- [S.2] **Soohwan Lee**, Seoyeong Hwang, and Kyungho Lee. (2025). **Beyond Individual UX: Defining Group Experience(GX) as a New Paradigm for Group-centered AI**. In Companion Publication of the 2025 ACM Designing Interactive Systems Conference (DIS'25 Companion)
- [S.3] **Soohwan Lee**, Seoyeong Hwang, Dajung Kim and Kyungho Lee. (2025). **Conversational Agents as Catalysts for Critical Thinking: Challenging Social Influence in Group Decision-making**. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI'25 Extended Abstract)
- [S.4] **Soohwan Lee**, Mingyu Kim, Seoyeong Hwang, Dajung Kim and Kyungho Lee. (2025). **Amplifying Minority Voices: AI-Mediated Devil's Advocate System for Inclusive Group Decision-Making**. In Companion Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25 Companion)
- [S.5] **Soohwan Lee**, and Kyungho Lee. (2023). **CheerUp: A Real-time Ambient Visualization of Cheerleading Pose Similarity**. In Companion Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23 Companion)

[Workshop Papers]

- [W.1] **Soohwan Lee**, Kyungho Lee (2026). **Delivered but Not Heard: AI Proxy Speech in Power-Imbalanced Group Decision-Making**. In Workshop on *Broader Impacts of GenAI in Communication: Building Agendas for Research, Design, and Policymaking* (CSCW '26 Workshop, Oral Presentation)
- [W.2] **Soohwan Lee**, Kyungho Lee (2026). **Where Does the Disagreement Go? Group Reflexivity and AI Accounts of Team Decision-makings**. In Workshop on *"Nobody Did This": Contribution, Originality, and Accountability in Agent-Mediated Collaboration* (CSCW '26 Workshop)
- [W.3] **Soohwan Lee**, Kyungho Lee (2026). **Multi-Agent Consensus as a Cognitive Bias Trigger in Human-AI Interaction**. In Bias4Trust Workshop on *Understanding, Mitigating, and Leveraging Cognitive Biases to Calibrate Trust in Evolving AI Systems* (CHI '26 Workshop)
- [W.4] **Soohwan Lee**, Kyungho Lee (2026). **Rethinking AI-Mediated Minority Support in Power-Imbalanced Group Decision-Making: From Anonymity To Authenticity**. In AI-MC Workshop on *Restoring Human Authenticity in AI-Mediated Communication* (CHI '26 Workshop)
- [W.5] **Soohwan Lee**, Seoyeong Hwang, and Kyungho Lee. (2024). **Conversational Agents as Catalysts for Critical Thinking: Challenging Design Fixation in Group Design**. In Workshop on *Death of the Design Researcher? Creating Knowledge Resources for Designers Using Generative AI* (DIS '24 Workshop, Oral Presentation)

[Thesis]

- [T.1] **Soohwan Lee**. (2022). **NEAS: AI-based Interactive System to Support Non-verbal Expressions in Online Communication**. Master Thesis, UNIST.

[Domestic Conference] (written in Korean)

- [D.1] **Soohwan Lee**, and Kyungho Lee. (2021). **NEAS: AI-based Interactive System to Support Non-verbal Expressions in Online Communication**. In Proceedings of 2021 Fall KSDS
- [D.2] Nari Kim, **Soohwan Lee**, Hyejin Oh, Temirlan Dzhorojev, and Young-Woo Park. (2021). **Interactive system design in everyday life to improve the perception of environmental hygiene information in pandemic situations**. In Proceedings of 2021 Spring KSDS
- [D.3] **Soohwan Lee**, and Kyungho Lee. (2021). **An Interactive Visualization of Gayageum's Expression and Mid-Air Gesture Recognition**. In Proceedings of 2021 Spring KSDS
- [D.4] **Soohwan Lee**, Kyungho Lee, and Young-Woo Park. (2021). **Counterfunctional Music Player with Mid-air Hand Gesture**. In Proceedings of HCI Korea 2021 (Creative Award)

[Under Review (TBA)]

- [U.1] **Soohwan Lee**, Kyungho Lee. TBA. **Reinforce, Align, or Oscillate? How People Experience and Interpret Multi-Agent Social Influence Over Time**. (Conference Paper)
- [U.2] **Soohwan Lee**, Seoyeong Hwang, and Kyungho Lee. TBA. **Reframing Supporting Minority Influence in the Age of AI: Bridging Theory, Design, and Simulation for Inclusive Group Decision-Making**. (Book Chapter)
- [U.3] **Soohwan Lee**, Soobin Lee and Kyungho Lee. TBA. **Expression2Emoji: Designing Non-verbal Communication Support Tool for Group Video Meetings**. (Conference Paper)

GRANTS

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|---|-----------------|
| • NRF (National Research Foundation of Korea) Grant for Doctoral Research
<i>PI, Total KRW 40M(approx. USD 30K), Accept Rate:31%, In Humanities and Social Sciences</i> | 09.2026–08.2028 |
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PROJECTS

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| • An Empirical Study on Supporting Reflection and User Experience in Group Decision-Making through the Visualization of Value Conflicts
<i>This work was supported by the Doctoral Student Research Fellowship(XXXXXXX) funded By the National Research Foundation of Korea</i> | 09/2026 - Present |
| • Development of a Smart CMF Recommendation and 3D Manufacturing Integration System
<i>This work was supported by the Industrial Technology Innovation Program(RS-2025-25457764, Smart CMF Design Recommendation and Real-Time Collaborative 3D Simulation System) funded By the Ministry of Trade, Industry & Energy(MOTIE, Korea)</i> | 10/2025 - Present |
| • Commercialization design and development of Intelligent Product-Service System for personalized full silver life cycle care
<i>This work was supported by the Technology Innovation Program (20015056, Commercialization design and development of Intelligent Product-Service System for personalized full silver life cycle care) funded By the Ministry of Trade, Industry & Energy(MOTIE, Republic of Korea).</i> | 01/2022 - 12/2025 |
| • Develop sensor modules and mountings for monitoring upper body angles and angular velocities
<i>This work was supported by LG Electronics.</i> | 01/2024 - 09/2024 |
| • Building an end-to-end speech recognition model for active seniors using KoSpeech model
<i>This research was supported by X-Corps Plus program of National Research Foundation of Korea(NRF) funded by the Ministry of Science, Republic of Korea, ICT (No. 2021H1D8A306520712).</i> | 03/2023 - 12/2023 |
| • Smart Clean BioAir Home Management System
<i>This work was supported by the Technology Innovation Program (20009977, Smart Clean BioAir Home Management System) funded By the Ministry of Trade, Industry & Energy(MOTIE, Republic of Korea).</i> | 04/2020 - 12/2021 |
| • UNIVERSE: Design of a Social Robot that Supports Emotional Well-Being of the MZ Gen
<i>This work was supported by Innovative Education Center, UNIST. (This project is part of AI Challenger Program at UNIST.)</i> | 09/2021 - 12/2021 |
| • Conceptualize and design a new UX/UI for a washer/dryer targeting Gen MZ.
<i>This work was supported by LG Electronics.</i> | 03/2021 - 09/2021 |

PATENT

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| • Air Shower Machine with Cradle
<i>KR 30-1146419-0000: This design combines the shape of the "air shower machine with a cradle" with the shape of the "air shower machine" as the main point of design creation.</i> | 12.2021 |
| • Clothes Rack with Mirror
<i>KR 30-1133312-0000: This design focuses on combining the shape and form of a "clothes rack with a mirror" as the main point of design creation.</i> | 10.2021 |

AWARDS & SCHOLARSHIPS

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| • UNIST Government-Funded Graduate Scholarship
<i>Full tuition waiver & annual stipend of KRW 10.6M–12.4M (approx. USD 7K–9K) during M.S. and Ph.D programs.</i> | 03.2020-02.2026 |
| • CES Innovation Awards
<i>2025 Honoree in Smart Home</i> | 02.2025
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TEACHING

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| • Teaching Assistant
<i>NOVA50601::Design for Agentic System and AI Transformation, UNIST</i> | 03/2026 - 06/2026 |
| • Teaching Assistant
<i>UNI207::Creative Computing for Emerging Media ART, UNIST</i> | 07/2022 - 08/2022 |

INVITED TALKS

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| • Collaborating with AI in Groups: A Design Research Approach to Social Influence
<i>Group Dynamics & Social Change Lab, Lehigh University (Hosted by Dr.Jiin Jung)</i> | 04/2025 |
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ACADEMIC SERVICES

[Organizing Committee]

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| • ACM IUI Student Volunteer Co-Chair | 2027 |
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[Program Committee]

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| • ACM CHI Poster Associate Chair | 2026 |
| • ACM CSCW Poster Associate Chair | 2026 |
| • ACM DIS PWiP Associate Chair | 2025 - 2026 |
| • ACM IUI Program Committee | 2026 - 2027 |

• ACM IUI HealthIUI Workshop Program Committee

2026

[Reviewer]

Have reviewed 50+ submissions in total, including full papers and posters.

6 Special Recognitions for Outstanding Reviews: CHI'26, IUI'26, UIST'25, DIS'26, DIS'25 Paper, CHI'25 LBW

- ACM CHI(26), CSCW(24), DIS(25-26), IUI(26), FAccT(26), UIST(25-26), C&C(26), AlpCHI(26) Paper Reviewer

2024 - 2026
- ACM CHI(25-26), DIS(25-26), CSCW(25-26), IUI(26), CUI(25) Poster & LBW & PWiP Reviewer

2025 - 2026
- International Journal of Human-computer Interaction Reviewer

2026

[Student Volunteer]

- ACM CHI Student Volunteer

2026
- ACM IUI Student Volunteer

2026

SKILLS

- **Research:** Qualitative & quantitative research, Experimental design, Mixed-methods, Statistical modeling, Behavioral analysis, Academic writing, Fast prototyping
- **Code & Systems:** Python (proficient), R (proficient), TypeScript, JavaScript, Node.js, HTML/CSS, LLM APIs, AI CLI tools (Codex, Claude Code, Antigravity, ETC.), Git, Processing, Arduino